

MH1200 Linear Algebra I.

Problem set #5.

SOLUTIONS

Problem 1:

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 & -3 \\ 1 & 1 & 0 & 4 \\ 0 & 4 & -1 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 0 & 1 & -3 \\ 2 & 2 & 0 & 8 \\ 0 & 4 & -1 & 2 \end{pmatrix}$$

$$\mathbf{C} = \begin{pmatrix} 0 & 4 & -1 & 2 \\ 1 & 1 & 0 & 4 \\ 2 & 0 & 1 & -3 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 & 1 & -3 \\ 1 & 1 & 0 & 4 \\ -1 & 3 & -1 & -2 \end{pmatrix}.$$

Note that

$$\mathbf{A} \xrightarrow{R_2 \rightarrow 2R_2} \mathbf{B}, \quad \mathbf{A} \xrightarrow{R_1 \leftrightarrow R_3} \mathbf{C}, \quad \mathbf{A} \xrightarrow{R_3 \rightarrow R_3 - R_2} \mathbf{D}.$$

1. If $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ are elementary matrices such that $\mathbf{E}_1\mathbf{A} = \mathbf{B}$, $\mathbf{E}_2\mathbf{A} = \mathbf{C}$, $\mathbf{E}_3\mathbf{A} = \mathbf{D}$, write down the matrices $\mathbf{E}_1, \mathbf{E}_2$ and $\mathbf{E}_3$. Check your answers.
2. Find elementary matrices $\mathbf{E}_4, \mathbf{E}_5$ and $\mathbf{E}_6$ such that $\mathbf{E}_4\mathbf{B} = \mathbf{A}$, $\mathbf{E}_5\mathbf{C} = \mathbf{A}$ and $\mathbf{E}_6\mathbf{D} = \mathbf{A}$. What elementary row operations do these three elementary matrices correspond to?
3. Verify that $\mathbf{E}_1\mathbf{E}_4 = \mathbf{E}_2\mathbf{E}_5 = \mathbf{E}_3\mathbf{E}_6 = \mathbf{I}$.

Solutions to 1.

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{E}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

Solutions to 2.

$$\mathbf{E}_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{E}_6 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

$\mathbf{E}_4$ corresponds to the row operation $R_2 \rightarrow \frac{1}{2}R_2$.

$\mathbf{E}_5$ corresponds to the row operation $R_1 \leftrightarrow R_3$.

$\mathbf{E}_6$ corresponds to the row operation $R_3 \rightarrow R_3 - R_2$.

Solutions to 3.

$$\mathbf{E}_1\mathbf{E}_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_2\mathbf{E}_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{E}_3\mathbf{E}_6 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

□

Problem 2:

Let $\mathbf{A}$ and $\mathbf{B}$ be $4 \times n$ matrices such that $\mathbf{E}_1\mathbf{E}_2\mathbf{A} = \mathbf{E}_3\mathbf{E}_4\mathbf{B}$ where

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$
$$\mathbf{E}_3 = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_4 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Describe how $\mathbf{B}$ can be obtained from $\mathbf{A}$ by elementary row operations.

Solution

$\mathbf{E}_1\mathbf{E}_2\mathbf{A} = \mathbf{E}_3\mathbf{E}_4\mathbf{B} \Rightarrow \mathbf{B} = \mathbf{E}_4^{-1}\mathbf{E}_3^{-1}\mathbf{E}_1\mathbf{E}_2\mathbf{A} \Rightarrow \mathbf{B}$ can be obtained from $\mathbf{A}$ by the following elementary row operations:

1. First, $R_2 \rightarrow R_2 - R_1$ (corresponds to multiplying by $\mathbf{E}_2$)
2. Next, $R_3 \rightarrow 2R_3$ (corresponds to multiplying by $\mathbf{E}_1$)
3. Next, $R_1 \rightarrow R_1 - 2R_3$ (corresponds to multiplying by $\mathbf{E}_3^{-1}$). This is the row operation which “undoes” the row operation corresponding to $\mathbf{E}_3$ which is $R_1 \rightarrow R_1 + 2R_3$.
4. Finally, $R_1 \leftrightarrow R_4$ (corresponds to multiplying by $\mathbf{E}_4^{-1}$). This is the row operation which “undoes” the row operation corresponding to $\mathbf{E}_4$ which is $R_1 \leftrightarrow R_4$, which in this case happens to be the same operation.

□

Problem 3:

(From the 2019 Midterm exam.)

Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} 0 & 2 & 3 \\ -1 & 2 & -2 \\ -1 & 4 & 1 \end{bmatrix}.$$

- (i) Give a sequence of elementary row operations which will transform this matrix into a matrix $\mathbf{B}$ in reduced row echelon form.
- (ii) Find elementary matrices $\mathbf{E}_1, \dots, \mathbf{E}_k$ such that

$$\mathbf{A} = \mathbf{E}_k \dots \mathbf{E}_1 \mathbf{B}.$$

- (iii) Using the information you deduced above, decide whether $\mathbf{A}$ is invertible or not. Briefly explain your deduction referring to any standard facts proved in the lectures you use.
- (iv) Let $\mathbf{C}$ and $\mathbf{D}$ be matrices with the property that they are row equivalent to the same row reduced echelon form. Do there exist elementary matrices $\mathbf{E}'_j$ such that

$$\mathbf{C} = \mathbf{E}'_m \dots \mathbf{E}'_1 \mathbf{D} \quad ?$$

Briefly justify.

Solution to (i).

We are asked to transform the matrix into **reduced** row echelon form. To do this we can follow Gauss-Jordan elimination.

$$\mathbf{A} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} -1 & 2 & -2 \\ 0 & 2 & 3 \\ -1 & 4 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} -1 & 2 & -2 \\ 0 & 2 & 3 \\ 0 & 2 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} -1 & 2 & -2 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

This part was Gaussian elimination and this matrix is now in row echelon form. To transform to reduced row echelon form we must continue:

$$\begin{array}{l} R_1 \rightarrow (-1)R_1 \\ R_2 \xrightarrow{\frac{1}{2}} R_2 \end{array} \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + 2R_2} \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 3/2 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{B}.$$

The sequence is as shown above.

Solution to (ii).

We are asked to find a sequence of elementary matrices such that

$$\mathbf{A} = \mathbf{E}_k \dots \mathbf{E}_1 \mathbf{B}.$$

There are a few different ways of organising this calculation. One way is to reverse the sequence of operations above. Note the sequence we just found goes in the other direction, from $\mathbf{A}$ to $\mathbf{B}$:

1. $R_1 \leftrightarrow R_2$
2. $R_3 \rightarrow R_3 - R_1$
3. $R_3 \rightarrow R_3 - R_2$
4. $R_1 \rightarrow (-1)R_1$
5. $R_2 \rightarrow \frac{1}{2}R_2$
6. $R_1 \rightarrow R_1 + 2R_2$

To go in the other direction, to get a sequence of operations that **starts** with $\mathbf{B}$ and **ends** with $\mathbf{A}$ we can take this sequence, start at the last operation and go backwards, at each step replacing an operation with the operation that “undoes” it. We get:

1. $R_1 \rightarrow R_1 - 2R_2$
2. $R_2 \rightarrow 2R_2$
3. $R_1 \rightarrow (-1)R_1$
4. $R_3 \rightarrow R_3 + R_2$
5. $R_3 \rightarrow R_3 + R_1$
6. $R_1 \leftrightarrow R_2$

The corresponding elementary matrices that implement these elementary row operations are obtained by performing the given operation on the 3×3 identity matrix. We get:

$$\mathbf{E}_1 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{E}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{E}_3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$\mathbf{E}_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \mathbf{E}_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \mathbf{E}_6 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Alternative approach to (ii).

The equation we would get directly from the sequence of operations we found by replacing the i -th row operations with an elementary matrix $\mathbf{E}'_i$ is:

$$\mathbf{B} = \mathbf{E}'_k \dots \mathbf{E}'_1 \mathbf{A}.$$

But notice that $\mathbf{B}$ and $\mathbf{A}$ are in the wrong places here! To get the equation we are asked for we have to take each matrix one-by-one to the other side of the equation. We do this by inverting the matrices, which are again elementary matrices. We get:

$$\mathbf{A} = (\mathbf{E}'_1)^{-1} \dots (\mathbf{E}'_k)^{-1} \mathbf{B}.$$

Then we would have to make sure we could take the inverses of the matrices correctly.

Solution to (iii).

One of the 4 characterisations of invertibility that we proved in the lectures is that a square matrix is invertible if and only if its reduced row echelon form is the identity matrix. But the reduced row echelon form we found above $\mathbf{B}$ is not the identity matrix. Thus we conclude that $\mathbf{A}$ is not invertible.

Solution to (iv).

Yes there do. If both $\mathbf{C}$ and $\mathbf{D}$ are row equivalent to the same row reduced echelon form then there are elementary matrices $\mathbf{E}_i$ and $\mathbf{F}_j$ such that

$$\mathbf{F}_n \dots \mathbf{F}_1 \mathbf{C} = \mathbf{E}_m \dots \mathbf{E}_1 \mathbf{D}.$$

If we multiply this equation on the left by the inverses of the $\mathbf{F}_i$'s then we deduce

$$\mathbf{C} = \mathbf{F}_1^{-1} \dots \mathbf{F}_n^{-1} \mathbf{E}_m \dots \mathbf{E}_1 \mathbf{D}.$$

Because we know that the inverses of elementary matrices are elementary matrices, the explanation is finished.

□

Problem 4

Let $\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ a & -1 & -2 \\ 2 & 3 & 7 \end{pmatrix}$ where a is an unknown constant.

1. Find the values of a for which $\mathbf{A}$ is invertible.

(Hint: One of the 4 characterisations of invertibility is that the corresponding homogeneous system $\mathbf{Ax} = \mathbf{0}$ only has the trivial solution. As always you can see if that is true by doing Gaussian elimination and seeing which values of a result in 1 solution.)

2. In the case of a such that $\mathbf{A}$ is invertible apply Gauss–Jordan Elimination to the 3×6 matrix $(\mathbf{A} | \mathbf{I})$ to determine its reduced row-echelon form. Use this to determine $\mathbf{A}^{-1}$. Check explicitly that your answer satisfies $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.

Solution

(Part 1.) This can be done by bringing $\mathbf{A}$ to row-echelon form by Gaussian Elimination:

$$\begin{aligned} \begin{pmatrix} 1 & 1 & 2 \\ a & -1 & -2 \\ 2 & 3 & 7 \end{pmatrix} &\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 2 \\ 2 & 3 & 7 \\ a & -1 & -2 \end{pmatrix} \xrightarrow{R_2 - 2R_1, R_3 - aR_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & -1-a & -2-2a \end{pmatrix} \\ &\xrightarrow{R_3 + (1+a)R_2} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1+a \end{pmatrix} \end{aligned}$$

Recall that the original square matrix $\mathbf{A}$ is invertible if and only if the row-echelon matrix has no zero row. In the present case this means $1+a \neq 0$, i.e. $a \neq -1$.

(Part 2.) We assume $a \neq -1$ so that $\mathbf{A}$ is invertible and apply Gauss–Jordan Elimination to $(\mathbf{A} | \mathbf{I})$:

$$\begin{aligned} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ a & -1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 7 & 0 & 0 & 1 \end{array} \right) &\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 2 & 3 & 7 & 0 & 0 & 1 \\ a & -1 & -2 & 0 & 1 & 0 \end{array} \right) \xrightarrow{R_2 - 2R_1, R_3 - aR_1} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 0 & 1 \\ 0 & -1-a & -2-2a & -a & 1 & 0 \end{array} \right) \\ &\xrightarrow{R_3 + (1+a)R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 0 & 1 \\ 0 & 0 & 1+a & -2-3a & 1 & 1+a \end{array} \right) \xrightarrow{\frac{1}{1+a}R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 0 & 1 \\ 0 & 0 & 1 & \frac{-2-3a}{1+a} & \frac{1}{1+a} & 1 \end{array} \right) \\ &\xrightarrow{R_1 - 2R_3, R_2 - 3R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & \frac{5+7a}{1+a} & \frac{-2}{1+a} & -2 \\ 0 & 1 & 0 & \frac{4+7a}{1+a} & \frac{-3}{1+a} & -2 \\ 0 & 0 & 1 & \frac{-2-3a}{1+a} & \frac{1}{1+a} & 1 \end{array} \right) \xrightarrow{R_1 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{1+a} & \frac{1}{1+a} & 0 \\ 0 & 1 & 0 & \frac{4+7a}{1+a} & \frac{-3}{1+a} & -2 \\ 0 & 0 & 1 & \frac{-2-3a}{1+a} & \frac{1}{1+a} & 1 \end{array} \right) \end{aligned}$$

By the algorithm for finding an inverse matrix we know that the final reduced row-echelon matrix is the same as $(\mathbf{I} | \mathbf{A}^{-1})$. Therefore:

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{1+a} & \frac{1}{1+a} & 0 \\ \frac{4+7a}{1+a} & \frac{-3}{1+a} & -2 \\ \frac{-2-3a}{1+a} & \frac{1}{1+a} & 1 \end{pmatrix}$$

Check $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$:

$$\mathbf{A}^{-1}\mathbf{A} = \begin{pmatrix} \frac{1}{1+a} & \frac{1}{1+a} & 0 \\ \frac{4+7a}{1+a} & \frac{-3}{1+a} & -2 \\ \frac{-2-3a}{1+a} & \frac{1}{1+a} & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 2 \\ a & -1 & -2 \\ 2 & 3 & 7 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

where the last equality follows from evaluation of the matrix product. For example, the $(2, 1)$ and $(2, 2)$ entries in the product matrix are found as follows:

$$\begin{aligned} (2, 1) \text{ entry: } & \frac{4+7a}{1+a} + \frac{-3}{1+a}a - 4 = \frac{4+7a-3a-4(1+a)}{1+a} = 0 \\ (2, 2) \text{ entry: } & \frac{4+7a}{1+a} + \frac{-3}{1+a}(-1) + (-2)3 = \frac{4+7a+3-6(1+a)}{1+a} = \frac{1+a}{1+a} = 1. \end{aligned}$$

□

Problem 5

Let $\mathbf{A} = \begin{pmatrix} a & b & c \\ 0 & b & c \\ 0 & 0 & c \end{pmatrix}$.

Determine the values of a, b, c for which $\mathbf{A}$ is invertible and under those conditions find $\mathbf{A}^{-1}$.

Solution

The easiest way to do this is to bring $\mathbf{A}$ to a simpler form by elementary row operations as follows:

$$\begin{pmatrix} a & b & c \\ 0 & b & c \\ 0 & 0 & c \end{pmatrix} \xrightarrow{R_1 - R_3} \begin{pmatrix} a & b & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

The last matrix is invertible precisely when a, b, c are all $\neq 0$ (since otherwise there is a zero row), so the same is true for $\mathbf{A}$.

To find $\mathbf{A}^{-1}$ we perform the same row operations as above on the extended matrix $(\mathbf{A} | \mathbf{I})$ and then bring it to reduced row-echelon form:

$$\begin{pmatrix} a & b & c & | & 1 & 0 & 0 \\ 0 & b & c & | & 0 & 1 & 0 \\ 0 & 0 & c & | & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 - R_3} \begin{pmatrix} a & b & 0 & | & 1 & 0 & -1 \\ 0 & b & 0 & | & 0 & 1 & -1 \\ 0 & 0 & c & | & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 - R_2} \begin{pmatrix} a & 0 & 0 & | & 1 & -1 & 0 \\ 0 & b & 0 & | & 0 & 1 & -1 \\ 0 & 0 & c & | & 0 & 0 & 1 \end{pmatrix}$$
$$\xrightarrow{\begin{matrix} \frac{1}{a}R_1 \\ \frac{1}{b}R_2 \\ \frac{1}{c}R_3 \end{matrix}} \begin{pmatrix} 1 & 0 & 0 & | & \frac{1}{a} & -\frac{1}{a} & 0 \\ 0 & 1 & 0 & | & 0 & \frac{1}{b} & -\frac{1}{b} \\ 0 & 0 & 1 & | & 0 & 0 & \frac{1}{c} \end{pmatrix}.$$

Since the reduced row-echelon matrix is also $(\mathbf{I} | \mathbf{A}^{-1})$, the inverse of $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{a} & -\frac{1}{a} & 0 \\ 0 & \frac{1}{b} & -\frac{1}{b} \\ 0 & 0 & \frac{1}{c} \end{pmatrix}$$

□

Problem 6

Give conditions on a, b, c, d for the following matrix to be invertible

$$\begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$$

(There is no need to find the inverse in this case.)

Solution

We need to decide for what values of the parameters the following system has exactly one solution:

$$\begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} \cdot$$

So as always we do Gaussian elimination:

$$\rightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix} \rightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix} \rightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-c \end{bmatrix}$$

This matrix has exactly one solution if and only if $a \neq 0$, $b \neq a$, $c \neq b$ and $d \neq c$.

□

Problem 7

Let $\mathbf{U}$ be any matrix satisfying the two properties that

- $\mathbf{U}$ is upper triangular
- every element on the diagonal is non-zero.

Briefly explain why $\mathbf{U}$ is invertible.

Solution

The augmented matrix of the corresponding linear system $\mathbf{U}\mathbf{x} = \mathbf{0}$ is already in row echelon form. Moreover, every column has a leading entry, so there is a unique solution. Thus by our characterisations of invertibility, the matrix $\mathbf{U}$ is invertible.

□

Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} a+1 & 1+a & 0 \\ a^2-1 & a^2 & 0 \\ 0 & 0 & a \end{bmatrix}.$$

- (a) For what values of the parameter a is $\mathbf{A}$ an invertible matrix? Justify.
- (b) For what values of the parameter a is $\mathbf{A}$ equal to a product of elementary matrices? Justify.
- (c) Assuming a is one of the values you gave in part (b), express $\mathbf{A}$ as a product of elementary matrices.

You may use standard results proved in theorems during the course in your deductions.

Solution to (a).

The determinant of this matrix is:

$$((a+1)a^2 - (1+a)(a^2-1))a = (a^3 + a^2 - (a^3 + a^2 - a - 1))a = a(a+1).$$

$\mathbf{A}$ will be invertible iff $\det(\mathbf{A}) \neq 0$. This will be true iff $a \neq 0, -1$.

Solution to (b).

We proved in lectures that being a product of elementary matrices is an alternative characterization of invertibility. Thus $\mathbf{A}$ will equal a product of elementary matrices if and only if $a \neq 0, -1$.

Solution to (c).

To express $\mathbf{A}$ as a product of elementary matrices, we need to know a sequence of elementary row operations that will takes us from the identity to $\mathbf{A}$. We obtain this sequence by taking the sequence occurring during Gauss-Jordan elimination then reversing it.

So we'll perform Gauss-Jordan elimination on $\mathbf{A}$, recording the elementary row operations we use.

$$\begin{bmatrix} a+1 & 1+a & 0 \\ a^2-1 & a^2 & 0 \\ 0 & 0 & a \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - (a-1)R_1} \begin{bmatrix} a+1 & 1+a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & a \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{1+a}R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & a \end{bmatrix}$$

Note this last move is valid because we are assuming $1+a \neq 0$. Also because $a \neq 0$ we can continue:

$$\xrightarrow{R_3 \rightarrow \frac{1}{a}R_3} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now to get a sequence of elementary row operation which will take us from $\mathbf{I}$ to $\mathbf{A}$, we reverse this sequence and replace each operation with its inverse. We get that $\mathbf{A}$ is obtained from $\mathbf{I}$ by

1. $R_1 \rightarrow R_1 + R_2$

2. $R_3 \rightarrow aR_3$
3. $R_1 \rightarrow (1+a)R_1$
4. $R_2 \rightarrow R_2 + (a-1)R_1$.

The expression we need follows by writing this sequence of operations as elementary matrices.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ a-1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1+a & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & a \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{I}$$

□

Problem 9:

(from midterm 2021.)

Consider the following two matrices:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 4 & 8 \\ 2 & 0 & 2 \\ 1 & -1 & -1 \end{bmatrix}.$$

- (i) Determine the row-reduced echelon form of these two matrices.
- (ii) Is it possible to find elementary matrices such that

$$\mathbf{A} = \mathbf{E}_k \dots \mathbf{E}_1 \mathbf{B} \quad ?$$

If it is indeed possible, determine such matrices. If it is not possible, explain why.

Solution to (i).

We determine RREF by performing Gauss-Jordan elimination.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

That is the RREF of $\mathbf{A}$. Next we do $\mathbf{B}$.

$$\begin{aligned} \mathbf{B} &= \begin{bmatrix} 0 & 4 & 8 \\ 2 & 0 & 2 \\ 1 & -1 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & -1 & -1 \\ 2 & 0 & 2 \\ 0 & 4 & 8 \end{bmatrix} \\ &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 4 \\ 0 & 4 & 8 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{bmatrix} \\ &\xrightarrow{R_2 \rightarrow 1/2 R_2} \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

Solution to (ii).

Notice that the two matrices have the same RREF. Thus they can be connected by a sequence of elementary row operations. The sequence of elementary row operations that starts with $\mathbf{B}$ and ends with $\mathbf{A}$ is obtained first listing the operations which take $\mathbf{B}$ to its RREF, and then listing the operations which go from the RREF back to $\mathbf{A}$. We obtain:

1. $R_1 \leftrightarrow R_3$
2. $R_2 \rightarrow R_2 - 2R_1$
3. $R_3 \rightarrow R_3 - 2R_2$
4. $R_2 \rightarrow 1/2R_2$
5. $R_1 \rightarrow R_1 + R_2$
6. $R_3 \rightarrow R_3 + R_2$
7. $R_3 \rightarrow R_3 + R_1$

Note that to get the last two operations we take the sequence which goes from $\mathbf{A}$ to its RREF, namely $R_3 \rightarrow R_3 - R_1$ and $R_3 \rightarrow R_3 - R_2$ and then reverse the sequence and replace the operation $R_3 \rightarrow R_3 - R_1$ by its inverse $R_3 \rightarrow R_3 + R_1$ and the operation $R_3 \rightarrow R_3 - R_2$ by its inverse $R_3 \rightarrow R_3 + R_2$.

Replacing this sequence with elementary matrices we write:

$$\mathbf{A} = \mathbf{E}_7\mathbf{E}_6\mathbf{E}_5\mathbf{E}_4\mathbf{E}_3\mathbf{E}_2\mathbf{E}_1\mathbf{B}$$

where

$$\mathbf{E}_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \mathbf{E}_2 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{E}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix},$$
$$\mathbf{E}_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{E}_5 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{E}_6 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \mathbf{E}_7 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

□

Problem 10:

(From the mid-term exam in 2022.)

Let $\mathbf{A}$ and $\mathbf{B}$ be invertible 3×3 matrices. Assume they are related by the equation

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{5}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{B}$$

Give a sequence of elementary row operations that transforms $\mathbf{A}$ to $\mathbf{B}$.

Solution

We rearrange this equation by multiplying on the left with inverses of elementary matrices to get:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \mathbf{A} = \mathbf{B}$$

Using the theory of elementary matrices, we can interpret this as a sequence of elementary row operations to deduce that $\mathbf{B}$ is obtained by applying the following sequence of elementary row operations to $\mathbf{A}$:

- $R_1 \leftrightarrow R_3$
- $R_1 \rightarrow R_1 - 2R_3$
- $R_3 \rightarrow R_3 - 1/2R_1$
- $R_2 \rightarrow 2/5R_2$.

Problem 11:

(From the 2022 midterm exam.)

Consider the following matrix $\mathbf{A}$ which depends on a single parameter $a \in \mathbb{R}$:

$$\mathbf{A} = \begin{bmatrix} 0 & a & a \\ a & a & a \\ a & a & 0 \end{bmatrix}.$$

Answer the following parts **without using determinants**.

- (i) Determine the values of a for which $\mathbf{A}$ is invertible.
- (ii) For such a , determine the inverse $\mathbf{A}^{-1}$.
- (iii) Does there exist an **invertible** 4×4 matrix $\mathbf{B}$ such that

$$\begin{bmatrix} 1 & 1 & 1 & 0 \end{bmatrix} \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 & 1 \end{bmatrix} \mathbf{B} ?$$

Justify.

- (iv) Is the following statement true or false: If $\mathbf{C} \in \mathbb{M}_{k \times k}(\mathbb{R})$ has the property that $\mathbf{C}^2 = \mathbf{0}_{k \times k}$, then $\mathbf{C} + \mathbf{I}_k$ is invertible. Justify.

Solution to (i).

There are different approaches to this.

According to our four equivalent characterisations of invertibility, this matrix is invertible if and only if the homogeneous system only has the trivial solution. We'll check that by seeing if we can perform Gaussian elimination, watching the parameter a .

$$\begin{bmatrix} 0 & a & a \\ a & a & a \\ a & a & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} a & a & a \\ 0 & a & a \\ a & a & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} a & a & a \\ 0 & a & a \\ 0 & 0 & -a \end{bmatrix}.$$

The corresponding homogeneous system only has the trivial solution if the parameter $a \neq 0$. (If $a = 0$ this is the zero matrix.)

Thus: $\mathbf{A}$ is invertible if and only if $a \neq 0$.

Solution to (ii).

Now we assume that $a \neq 0$ and proceed to calculate the inverse according to the algorithm we learnt in the lectures.

$$\left[\begin{array}{ccc|ccc} 0 & a & a & 1 & 0 & 0 \\ a & a & a & 0 & 1 & 0 \\ a & a & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} a & a & a & 0 & 1 & 0 \\ 0 & a & a & 1 & 0 & 0 \\ a & a & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|ccc} a & a & a & 0 & 1 & 0 \\ 0 & a & a & 1 & 0 & 0 \\ 0 & 0 & -a & 0 & -1 & 1 \end{array} \right]$$

$$\begin{array}{l} R_1 \rightarrow \frac{1}{a}R_1 \\ R_2 \rightarrow \frac{1}{a}R_2 \\ R_3 \rightarrow -\frac{1}{a}R_3 \end{array} \xrightarrow{\quad} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1/a & 0 \\ 0 & 1 & 1 & 1/a & 0 & 0 \\ 0 & 0 & 1 & 0 & 1/a & -1/a \end{array} \right] \xrightarrow{\begin{array}{l} R_2 \rightarrow R_2 - R_3 \\ R_1 \rightarrow R_1 - R_3 \end{array}} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 0 & 1/a \\ 0 & 1 & 0 & 1/a & -1/a & 1/a \\ 0 & 0 & 1 & 0 & 1/a & -1/a \end{array} \right]$$

$$\xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1/a & 1/a & 0 \\ 0 & 1 & 0 & 1/a & -1/a & 1/a \\ 0 & 0 & 1 & 0 & 1/a & -1/a \end{array} \right].$$

The inverse is thus

$$\mathbf{A}^{-1} = \begin{bmatrix} -1/a & 1/a & 0 \\ 1/a & -1/a & 1/a \\ 0 & 1/a & -1/a \end{bmatrix}$$

Solution to (iii).

There cannot exist such an invertible matrix $\mathbf{B}$. If there was such a matrix, then we could multiply this equation on the right by $\mathbf{B}^{-1}$ to deduce:

$$\begin{bmatrix} 1 & 1 & 1 & 0 \end{bmatrix} \mathbf{B} \mathbf{B}^{-1} = \begin{bmatrix} 1 & 1 & 0 & 1 \end{bmatrix} \mathbf{B} \mathbf{B}^{-1}$$

Because the $\mathbf{B}$ cancels the $\mathbf{B}^{-1}$ this would imply that

$$\begin{bmatrix} 1 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 \end{bmatrix}.$$

But this equation isn't true. So an invertible $\mathbf{B}$ solving this equation can't exist. Note that this problem tests understanding of part of the proof of the 4 equivalent characterisations of invertibility.

Solution to (iv).

The statement is **true**. If $\mathbf{C} \in \mathbb{M}_{k \times k}(\mathbb{R})$ has the property that $\mathbf{C}^2 = \mathbf{0}_{k \times k}$, then observe that

$$(\mathbf{C} + \mathbf{I}_k)(-\mathbf{C} + \mathbf{I}_k) = -\mathbf{C}^2 - \mathbf{I}_k \mathbf{C} + \mathbf{C} \mathbf{I}_k + \mathbf{I}_k \mathbf{I}_k = \mathbf{I}_k.$$

Also

$$(-\mathbf{C} + \mathbf{I}_k)(\mathbf{C} + \mathbf{I}_k) = -\mathbf{C}^2 + \mathbf{I}_k \mathbf{C} - \mathbf{C} \mathbf{I}_k + \mathbf{I}_k \mathbf{I}_k = \mathbf{I}_k.$$

Thus $\mathbf{C} + \mathbf{I}_k$ has inverse $-\mathbf{C} + \mathbf{I}_k$.

Alternative solution to (iv).

Some students suggested the following alternative solution.

We can use the theorem about the 4 characterisations of invertibility to show the matrix $\mathbf{C} + \mathbf{I}_k$ is invertible by proving that the only column vector $\mathbf{x}$ solving the equation

$$(\mathbf{C} + \mathbf{I}_k) \mathbf{x} = \mathbf{0}_{k \times 1} \quad (\star).$$

is the zero vector $\mathbf{0}_{k \times 1}$.

Assume $\mathbf{x}$ solves this equation. If we multiply both sides of this equation by $\mathbf{C}$ we deduce the equation:

$$(\mathbf{C}^2 + \mathbf{C}) \mathbf{x} = \mathbf{0}_{k \times 1}.$$

Using the fact that $\mathbf{C}^2 = \mathbf{0}_{k \times k}$ and the fact that $\mathbf{C}\mathbf{x} = -\mathbf{x}$ (which follows from $(\star)$) this implies that

$$\mathbf{x} = \mathbf{0}_{k \times 1}.$$

Thus by the big theorem characterising invertibility we deduce that $\mathbf{C} + \mathbf{I}_k$ is invertible.

□

More abstract or challenging problems:

Problem 12:

Consider the following matrix:

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

What is the inverse of this matrix?

Now extend $\mathbf{A}$ to a general $n \times n$ matrix A_n by continuing this pattern of 1's and -1 's. Can you write down the inverse of the general case $\mathbf{A}_n^{-1}$? Give some justification.

Solution

This matrix is already in row echelon form. Clearly every column contains the leading entry of some row, so the system has a unique solution, so the matrix is invertible.

The inverse is calculated in the usual way: by putting an identity matrix next to this matrix, then doing Gauss-Jordan elimination to turn the first half into an identity matrix.

The key observation here is that the sequence of elementary row operations we use is going to be very simple and predictable:

$$R_1 \rightarrow R_1 + R_2, \quad R_2 \rightarrow R_2 + R_3, \quad R_3 \rightarrow R_3 + R_4.$$

Observe:

$$\mathbf{A} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

If we apply the same sequence of elementary row operations to the identity matrix we calculate the inverse:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Thus the inverse of the given $\mathbf{A}$ is:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Now we think about what happens in general. For example the 8×8 case of this matrix is:

$$\mathbf{A}_8 = \begin{bmatrix} 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The trick to note is that for each row, the part after the leading entry is exactly (-1) times the next row. So if we add the next row to it, we kill the whole row except the leading entry which is 1. So the sequence of elementary row operations which puts this matrix into RREF is:

$$R_1 \rightarrow R_1 + R_2, R_2 \rightarrow R_2 + R_3, \dots, R_7 \rightarrow R_7 + R_8.$$

Is we apply this sequence to the identity matrix we obviously get: the identity matrix but in addition every entry above the diagonal is changed to 1.

$$\mathbf{A}_8^{-1} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Problem 13:

Let $\mathbf{A}$ and $\mathbf{B}$ be two square matrices of the same order. Prove that if $\mathbf{A}$ is singular (i.e. not invertible) then $\mathbf{AB}$ and $\mathbf{BA}$ are also both singular.

Hint: Go back and have a look at our 4 characterisations of invertibility. If you use one of these 4 characterisations in the right way you can give a simple proof of this property.

Solution

We will base our proof on the following two properties:

1. A square matrix $\mathbf{A}$ is invertible if and only if the matrix equation $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution. (This was proved in lectures.)
2. A square matrix $\mathbf{A}$ is invertible if and only if its transpose $\mathbf{A}^T$ is invertible. (This was proved in an earlier tutorial.)

Assume that $\mathbf{A}$ is not invertible.

We will first prove that $\mathbf{BA}$ is not invertible. Because $\mathbf{A}$ is not invertible it follows from Property 1 that there is a non-zero solution $\mathbf{x}$ of the equation $\mathbf{Ax} = \mathbf{0}$. This implies there is a non-zero solution of the equation $\mathbf{BAx} = \mathbf{0}$. This implies (again by Property 1 above) that the matrix $\mathbf{BA}$ is not invertible, which is what we wanted to establish.

Next we need to prove that $\mathbf{AB}$ is also not invertible. Assume on the other hand that it is invertible. Property 2 above would then imply that $(\mathbf{AB})^T$ is invertible, which would imply that $\mathbf{B}^T\mathbf{A}^T$ is invertible, which would imply that $\mathbf{A}^T$ is invertible because of the first part of this problem. By a second use of Property 2 this would imply that $\mathbf{A}$ is invertible, which contradicts the assumption about $\mathbf{A}$.

□

Problem 14

(From final exam 2020.)

Let $R(\mathbf{A})$ denote the result of applying the elementary row operation R which swaps two rows i and j of a matrix $\mathbf{A}$, where $\mathbf{A}$ is some matrix with p rows and $1 \leq i \neq j \leq p$. Let $\mathbf{E}$ denote the corresponding elementary matrix.

(a) Prove that for any $p \times q$ matrix $\mathbf{A}$ and any $q \times r$ matrix $\mathbf{B}$

$$R(\mathbf{AB}) = R(\mathbf{A})\mathbf{B}.$$

(b) Using Part (a), prove the fundamental fact for this elementary operation. Namely, show:

$$\mathbf{EA} = R(\mathbf{A})$$

for every matrix $\mathbf{A}$ with p rows.

Proof of (a)

There are different ways of organising this simple explanation. The way that is closest to the discussion in the lecture notes is: Organize $\mathbf{A}$ into rows and organize $\mathbf{B}$ into columns as follows:

$$\mathbf{AB} = \begin{bmatrix} \leftarrow & A_1 & \rightarrow \\ \leftarrow & A_2 & \rightarrow \\ & \vdots & \\ \leftarrow & A_p & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ B_1 & B_2 & \cdots & B_r \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} = \begin{bmatrix} A_1B_1 & \cdots & A_1B_r \\ & \vdots & \\ A_iB_1 & \cdots & A_iB_r \\ & \vdots & \\ A_jB_1 & \cdots & A_jB_r \\ & \vdots & \\ A_pB_1 & \cdots & A_pB_r \end{bmatrix}$$

Now we turn to the identity we are proving. The LHS $R(\mathbf{AB})$ equals:

$$\begin{bmatrix} A_1B_1 & \cdots & A_1B_r \\ & \vdots & \\ A_jB_1 & \cdots & A_jB_r \\ & \vdots & \\ A_iB_1 & \cdots & A_iB_r \\ & \vdots & \\ A_pB_1 & \cdots & A_pB_r \end{bmatrix}.$$

This equals:

$$\begin{bmatrix} \leftarrow & A_1 & \rightarrow \\ & \vdots & \\ \leftarrow & A_j & \rightarrow \\ & \vdots & \\ \leftarrow & A_i & \rightarrow \\ & \vdots & \\ \leftarrow & A_p & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ B_1 & B_2 & \cdots & B_r \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

This equals $R(\mathbf{A})\mathbf{B}$ and we are done.

Alternative proof of (a)

We are proving $R(\mathbf{AB}) = R(\mathbf{A})\mathbf{B}$.

The (k, l) -entry of the LHS is:

$$\begin{cases} \text{The } (j, l)\text{-entry of } \mathbf{AB} & \text{if } k = i \\ \text{The } (i, l)\text{-entry of } \mathbf{AB} & \text{if } k = j \\ \text{The } (k, l)\text{-entry of } \mathbf{AB} & \text{otherwise.} \end{cases} = \begin{cases} \sum_{x=1}^q A_{jx}B_{xl} & \text{if } k = i \\ \sum_{x=1}^q A_{ix}B_{xl} & \text{if } k = j \\ \sum_{x=1}^q A_{kx}B_{xl} & \text{otherwise.} \end{cases}$$

The (k, l) -entry of the RHS equals:

$$\sum_{x=1}^q (R(\mathbf{A}))_{kx} \mathbf{B}_{xl}.$$

This equals:

$$\begin{cases} \sum_{x=1}^q A_{jx}B_{xl} & \text{if } k = i \\ \sum_{x=1}^q A_{ix}B_{xl} & \text{if } k = j \\ \sum_{x=1}^q A_{kx}B_{xl} & \text{otherwise.} \end{cases}$$

The two sides are equal so the proof is complete.

Proof of (b)

RHS equals

$$R(\mathbf{A}) = R(\mathbf{IA}) = R(\mathbf{I})\mathbf{A} = \mathbf{EI}.$$

The second equality uses the fact proved in part (a).

□

Problem 15

Consider a 4×4 matrix $\mathbf{A}$ where each column has exactly the numbers 0, 1, 2, and -3 in some unspecified order. Is it possible for such a matrix to be invertible? Justify your answer.

Solution

No it's not possible.

First we explain that the transpose of $\mathbf{A}$ cannot be invertible. Well, the transpose of $\mathbf{A}$ is a matrix which has the numbers 0, 1, 2 and -3 in every row. It is easy to see that then

$$\mathbf{A}^T \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

(This is because the entries of the RHS are just the sum of the entries of the corresponding row: $0 + 1 + 2 + -3 = 0$.)

Thus $\mathbf{A}^T$ is not invertible. Thus $\mathbf{A}$ is also not invertible.

□

Problem 16:

Let $n \in \mathbb{N}$ and let i and j be different numbers such that $1 \leq i \leq n$ and $1 \leq j \leq n$. Let $c \in \mathbb{R}$.

Let $\mathbf{E}$ be the $n \times n$ elementary matrix corresponding to the elementary row operation $R_i \rightarrow R_i + cR_j$.

Let $\mathbf{M}$ be a $n \times n$ matrix.

Explain how to obtain the matrix $\mathbf{ME}$ from $\mathbf{M}$ by means of a **column** operation. Justify your answer.

Solution

Let's work it out by turning it into a row operation using transpose. (Note there will be many other ways to do this, such as by understanding the product using a "block" decomposition. Some sort of justification is necessary to get full marks.)

First note the matrix $\mathbf{E}$ is the identity matrix with an extra c in the (i, j) position.

Now note that because doing transpose twice has no effect, we can write:

$$\mathbf{ME} = (\mathbf{E}^T \mathbf{M}^T)^T.$$

Let's identify $\mathbf{E}^T$. It is just the identity matrix with a single extra c in the (j, i) position. This is precisely the elementary matrix corresponding to the elementary row operation $R_j \rightarrow R_j + cR_i$. So by the theory of elementary matrices we deduce that $\mathbf{ME}$ equals

$$(\text{The result of applying } R_j \rightarrow R_j + cR_i \text{ to } \mathbf{M}^T)^T.$$

When we take transpose, rows become columns and columns become rows. So this is precisely the result of applying the column operation

$$C_j \rightarrow C_j + cC_i$$

to $\mathbf{M}$.

□

Let's check our answer using the example of $R_2 \rightarrow R_2 - 3R_3$:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

The product is:

$$\begin{bmatrix} 1 & 2 & -3 \\ 4 & 5 & -9 \\ 7 & 8 & -15 \end{bmatrix}.$$

As predicted, this is obtained by applying $C_3 \rightarrow C_3 - 3C_2$ to $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.